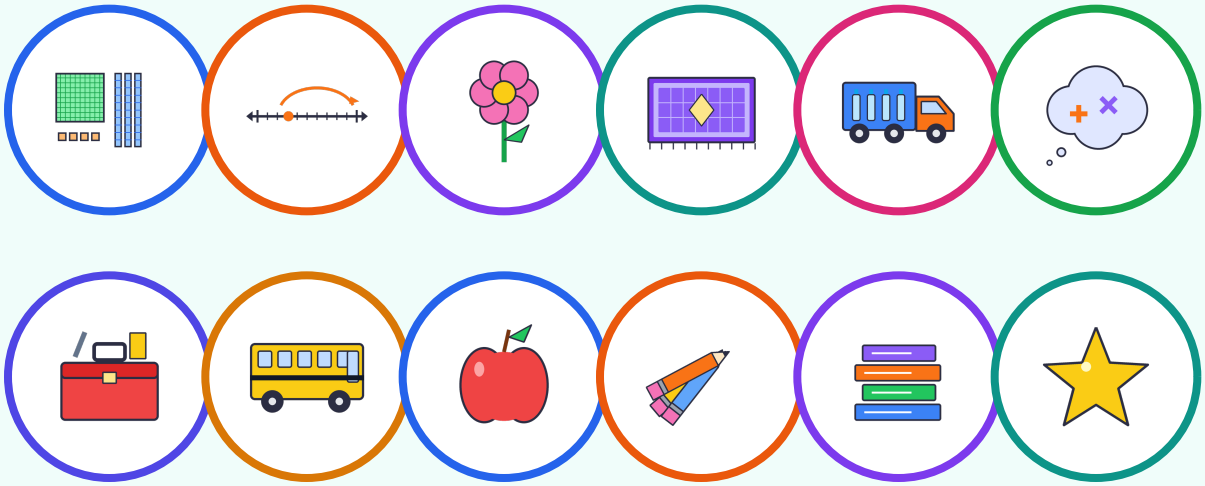
1 2 3 4 5 6 7 8

GRADE 4 MATH WORKBOOK

Multiplying by One-Digit Numbers

Step-by-step lessons, guided practice, and notes for educators



This book belongs to _____

Educator's Guide

For the teacher or parent working alongside the student.

WHAT THIS BOOK TEACHES

Students learn to multiply two-, three-, and four-digit numbers by a one-digit number: zero patterns, estimating, breaking numbers apart by place value, mental math, choosing a strategy, and bar models for word problems.

How each lesson works

	**Page 1: Learn It** Read it together. A short note, key words, and one example worked step by step.	**Together** about 10 min
	**Page 2: Try It Together** Guided problems. The first ones fill in most steps; each one after gives a little less help.	**Together** 10 to 15 min
	**Page 3: Practice** The student works alone with the hints removed, then solves a word problem.	**Alone** about 15 min
	**Page 4: More Practice** A skill builder, two more word problems, and a mixed review of earlier lessons.	**Alone** about 15 min
	**Page 5: Check Yourself** A short check, a find-the-mistake item, an SAT Connection, and a self-rating.	**Alone** 5 to 10 min

A GOOD ROUTINE

Spread each lesson over three short sessions.

Day 1: Learn It and Try It Together.

Day 2: Practice.

Day 3: More Practice and Check Yourself.

About six weeks at four sessions a week.

YOU WILL NEED

A pencil and an eraser

Colored pencils for the self-rating

Scrap paper for drawing

Place-value blocks (optional)



THE ESTIMATE HABIT From Lesson 2 on, students estimate first. Afterward ask: *"Should your answer be more or less than your estimate? Why?"* Rounded up means the exact answer is less; rounded down means it is more. Close to an estimate does not prove an answer right, but it catches answers far too big or small.

Educator's Guide: How to Help

Try saying this	Instead of this
"What basic fact do you see?"	"That's wrong."
"Which place is that digit in? What is it worth?"	"You forgot a zero."
"Show me your estimate first."	"Just do it the fast way."
"Is your answer more or less than your estimate? Why?"	"Is that right?"
"Can you draw it?"	Doing the problem for the student

WHEN THE STUDENT IS STUCK

1. Go back to the example on the Learn It page.
2. Open the Student Quick Guide at the front of the book.
3. Draw blocks, an array, or a rectangle.
4. Try the same problem with smaller numbers first.

USING THE ANSWER KEY

The answer key is at the back of the book.

Check each page after it is finished. Circle mistakes, then let the student find and fix them.

Accept other reasonable estimates and any correct strategy that is shown.

WHAT THE SELF-RATING MEANS

At the end of each lesson the student colors one circle. Check the student's answers and ask them to explain their method. Move on when they demonstrate understanding; use the self-rating to discuss confidence and support needs.

STANDARDS

This book supports Virginia Mathematics Standards of Learning (2023) 4.CE.2 and Common Core State Standard 4.NBT.B.5.

Lesson 5 (four-digit factors) and the four-digit problems in later lessons go beyond Virginia's Grade 4 computation limits (two-digit and three-digit by one-digit). They are part of Common Core 4.NBT.B.5. Treat them as a stretch lesson in Virginia classrooms.

LESSON MAP

Lesson 1 Zero Patterns **Tue Oct 13**
Lesson 2 Estimating **Wed Oct 14**
Lesson 3 Arrays **Thu Oct 15**
Lesson 4 Area Models **Fri Oct 16**
Lesson 5 Four Digits (stretch) **Mon Oct 19**
Lesson 6 Mental Math **Tue Oct 20**
Lesson 7 Pick Your Tool **Wed Oct 21**
Lesson 8 Bar Models **Thu Oct 22**

The Big Idea

A short explanation of the topic for students and educators.

What does it mean to multiply?

Multiplying means finding the total of **equal groups**. 3 × 24 means 3 groups of 24.

Big numbers can be hard to multiply all at once. So we use **place value** to break a big number into friendly parts: 24 is 2 tens and 4 ones. We multiply each part, then put the parts back together.

One problem, three ways: 3 × 24

Equal groups

24 + 24 + 24 = 72

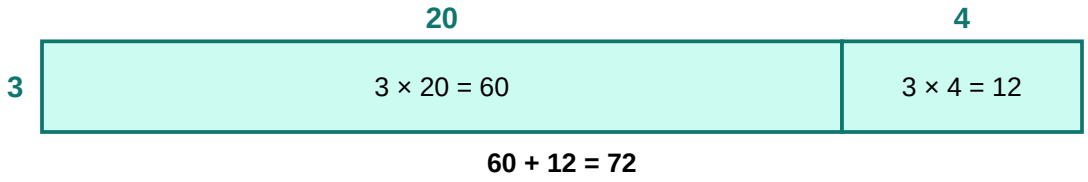
Adding works, but it is slow for big numbers.

Break it apart

3 × 20 = 60
3 × 4 = 12
60 + 12 = 72

Estimate first

24 is about 20.
3 × 20 = 60.
72 is a little more, because 24 was rounded down.



Your learning path

1 Zero patterns
2 Estimate
3 Arrays
4 Area models
5 Four digits
6 Mental math
7 Pick a tool
8 Bar models

THREE HABITS OF A STRONG MATH THINKER

1. **Estimate first.** Know about how big the answer should be.
2. **Show every part.** Write one line for each partial product.
3. **Check the size.** Ask: should my answer be more or less than my estimate?

SAT CONNECTIONS Each lesson ends with an SAT Connection box. It shows how the skill grows into an idea that high school students use on the math section of the SAT. This book teaches Grade 4 skills; it is not SAT test preparation.

SAT is a registered trademark of the College Board, which is not affiliated with and does not endorse this product.

Student Quick Guide

Keep this page open while you work. Each card matches a lesson.

1 Zero Patterns

1. Cover the zeros.
2. Do the basic fact.
3. Put the zeros back.

Example 4 × 300: 4 × 3 = 12, so 1,200

2 Estimate

1. Round the big number to its greatest place.
2. Multiply.
3. Write "about."

Example 3 × 289: 289 → 300. About 900

3 Partial Products

1. Break the big number by place value.
2. Multiply each part.
3. Add the parts.

Example 3 × 24 = 60 + 12 = 72

4 Area Model

1. Draw a rectangle and split it by place value.
2. Write each part's product inside.
3. Add.

Example 5 × 38: 150 + 40 = 190

PLACE VALUE HELPER

Thousands	Hundreds	Tens	Ones
3	2	1	4
3,000	200	10	4

3,214 = 3,000 + 200 + 10 + 4. Each digit is worth its place.

Student Quick Guide (continued)

5 Four Digits



1. Write one line for each place.
2. Thousands, hundreds, tens, ones.
3. Add.

Example 2 × 3,214 = 6,000 + 400 + 20 + 8 = 6,428

6 Mental Math



1. Look for 4 × 25 or 2 × 50.
2. Or use a friendly number like 200.
3. Fix the extra: add or subtract.

Example 5 × 199: 1,000 − 5 = 995

7 Pick Your Tool



1. Zeros? Use the zero pattern.
2. Close to friendly? Make it friendly.
3. Otherwise? Break it apart.

Example 4 × 98: 400 − 8 = 392

8 Bar Models



1. Draw one box for each group.
2. Write the number in each box.
3. Write an equation. Solve.

Example 5 buses of 42: 5 × 42 = 210



MORE OR LESS?

Rounded **up**? Your exact answer is **less** than the estimate.

Rounded **down**? Your exact answer is **more** than the estimate.



FIX THE EXTRA

Friendly number **bigger** than yours? **Subtract** the extra.

Friendly number **smaller** than yours? **Add** what is missing.

LESSON 1

Zero Patterns

Tuesday, October 13

Name _____ Date _____

BIG IDEA To multiply by a number like 30, 300, or 3,000, use a basic fact and place value.

WORDS TO KNOW

factor a number you multiply	**product** the answer	**basic fact** like 4 × 3 = 12	**place value** what a digit is worth

LEARN IT — Watch the pattern grow

Problem	Think	Product
4 × 3	4 groups of 3 ones = 12 ones	**12**
4 × 30	4 groups of 3 **tens** = 12 tens	**120**
4 × 300	4 groups of 3 **hundreds** = 12 hundreds	**1,200**
4 × 3,000	4 groups of 3 **thousands** = 12 thousands	**12,000**

THE ZERO SHORTCUT

1. Cover the zeros. Do the basic fact. 6 × 4<u>00</u>: 6 × 4 = 24
2. Count the zeros you covered. 400 has 2 zeros.
3. Write them after the basic fact answer. 6 × 400 = 2,400

WATCH OUT! Some basic facts already end in zero. 5 × 400: 5 × 4 = **20**. Add two more zeros: **2,000**. Keep the zero in 20!

SAY IT OUT LOUD "7 times 2 hundreds is 14 hundreds, which is 1,400." Saying the place helps you see why the zeros are there.

TRY IT TOGETHER **Step by step. Each problem gives a little less help.**

1. 3 × 40

Cover the zero: 3 × 4 = _____

Zeros covered: _____

3 × 40 = _____

2. 6 × 500

Basic fact: 6 × _____ = _____

Zeros covered: _____

6 × 500 = _____

3. Finish the pattern. The first row is done for you.

Basic fact	× 10	× 100	× 1,000
2 × 8 = 16	2 × 80 = 160	2 × 800 = 1,600	2 × 8,000 = 16,000
7 × 3 = _____	7 × 30 = _____	7 × 300 = _____	7 × 3,000 = _____
5 × 6 = _____	5 × 60 = _____	5 × 600 = _____	5 × 6,000 = _____

4. 8 × 2,000

Basic fact: _____ × _____ = _____

Zeros covered: _____

8 × 2,000 = _____

5. 4 × 500

Basic fact: _____ × _____ = _____

Does the basic fact end in zero? YES NO

4 × 500 = _____

6. 9 × 70

9 × 70 = _____

7. 3 × 6,000

3 × 6,000 = _____

8. 6 × 50

6 × 50 = _____

EDUCATOR TIP Ask the student to say the basic fact and the place out loud ("9 times 7 tens is 63 tens") before writing any zeros.

PRACTICE On your own

Use the zero shortcut. Watch for basic facts that already end in zero.

a) 4 × 60 = _____	b) 7 × 300 = _____
c) 2 × 9,000 = _____	d) 5 × 800 = _____
e) 8 × 40 = _____	f) 6 × 7,000 = _____
g) 4 × 5,000 = _____	h) 9 × 600 = _____

PRACTICE Word problem

9. A school orders 7 boxes of pencils. Each box has 200 pencils. How many pencils is that in all?

Basic fact: _____ × _____ = _____ Zeros: _____

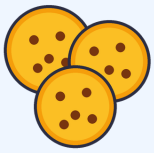
Answer: _____ pencils



10. A bakery sells 3,000 muffins each month. How many muffins does it sell in 4 months?

_____ × _____ = _____

Answer: _____ muffins



PRACTICE Pattern detective

11. Mia says, "When I multiply a positive whole number by 100, the product always ends in exactly two zeros." Is she right? Try 10 × 100.

10 × 100 = _____ Is Mia right? YES NO

Explain:

_____

_____

MORE PRACTICE Skill builder

Use the zero shortcut.

a) 3 × 50 = _____	b) 8 × 700 = _____	c) 6 × 4,000 = _____
d) 2 × 90 = _____	e) 7 × 600 = _____	f) 5 × 200 = _____
g) 9 × 3,000 = _____	h) 4 × 800 = _____	i) 8 × 60 = _____
j) 6 × 900 = _____	k) 5 × 4,000 = _____	l) 3 × 7,000 = _____

MORE PRACTICE Word problems

1. A farm packs 6 crates. Each crate holds 300 apples. How many apples is that?

_____

_____

Answer: _____ apples

2. A bus trip is 40 miles. How many miles are 9 trips?

_____

_____

Answer: _____ miles

CHECK YOURSELF **Show what you know**

1. 9 × 40

= _____

2. 3 × 800

= _____

3. 6 × 5,000

= _____

4. Find the mistake. Sam says 4 × 500 = 200.

The correct answer is _____.

Explain how to correct Sam's answer.

_____

_____

5. Make your own. Write a problem with the product 2,400. Use a one-digit number and a number that ends in zeros.

_____ × _____ = 2,400

SAT CONNECTION

IF YOU MASTER
Multiplying by tens, hundreds, and thousands

→

IT HELPS YOU UNDERSTAND
powers of ten, place value, and changing units

→

ON THE SAT
Problem-Solving and Data Analysis: unit conversions and rates

Example: 4 kilometers = 4 × 1,000 = 4,000 meters

How did I do? Color one. I can do this alone · I need a little help · I need more practice

EDUCATOR TIP Check the student's answers and ask them to explain their method. Move on when they demonstrate understanding; use the self-rating to discuss confidence and support needs. For extra practice, redo Try It Together problems 4 and 5 with new numbers, such as 6 × 500 and 8 × 50.

LESSON 2

About How Much? Estimating Products

Wednesday, October 14

Name _____ Date _____

BIG IDEA An estimate is a close answer you can find fast. It tells you about how big the exact answer should be.

WORDS TO KNOW

estimate a close answer	**round** change to a nearby friendly number	**about** the word for an estimate	**greatest place** the biggest place in a number

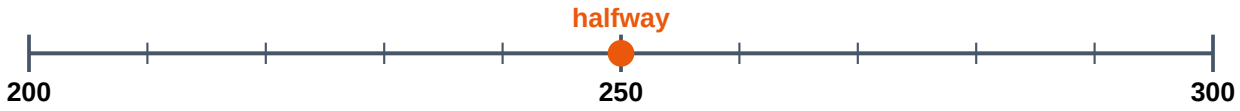
LEARN IT How to round

FOUR ROUNDING STEPS

1. Find the rounding place. Use the greatest place: tens for 2 digits, hundreds for 3 digits, thousands for 4 digits.
2. Look at the digit just to the **right** of the rounding place.
3. **0, 1, 2, 3, 4:** round down (the rounding digit stays the same). **5, 6, 7, 8, 9:** round up (add 1 to the rounding digit).
4. Change every digit to the right of the rounding place to zero.

Number	Rounding place	Digit to the right	Round	Rounded
58	tens (5)	8	up	60
389	hundreds (3)	8	up	400
250	hundreds (2)	5	up	300
2,430	thousands (2)	4	down	2,000

250 is exactly halfway between 200 and 300. Halfway rounds **up**, so 250 rounds to **300**.



WORKED EXAMPLE: 4 × 389

1. Round the big number to its greatest place. 389 → 400
2. Multiply. 4 × 400 = 1,600
3. Write "about." 4 × 389 is **about 1,600**.

389 was rounded **up**, so the exact answer will be a little **less** than 1,600. Round the big number only: the 4 stays a 4.

TRY IT TOGETHER — Step by step

1. Is 58 closer to 50 or 60? Circle the answer. 50 60



2. 3 × 58

58 rounds to _____.

3 × _____ = _____

3 × 58 is about _____.

3. 6 × 712

712 has three digits. Round to the nearest **hundred**.

712 rounds to _____.

6 × _____ = _____

About _____

4. 5 × 2,830

Greatest place: _____

Rounded factor: _____

Multiply: 5 × _____ = _____

About _____

5. 7 × 41

Rounded factor: _____

Multiply: _____ × _____ = _____

About _____

Rounded up or down? UP DOWN

6. More or less? In problem 5, you rounded 41. Will the exact answer be more or less than your estimate? MORE LESS

Why?

_____

_____

EDUCATOR TIP Keep the number line handy. If the student hesitates, ask which two tens (or hundreds) the number sits between, then which one is closer.

PRACTICE On your own

Round the big number to its greatest place. Multiply. Write your estimate.

a) 7 × 42

Rounded factor: _____

Multiply: _____ × _____ = _____

Estimate: about _____

b) 4 × 67

Rounded factor: _____

Multiply: _____ × _____ = _____

Estimate: about _____

c) 8 × 315

Rounded factor: _____

Multiply: _____ × _____ = _____

Estimate: about _____

d) 3 × 586

Rounded factor: _____

Multiply: _____ × _____ = _____

Estimate: about _____

e) 6 × 1,920

Rounded factor: _____

Multiply: _____ × _____ = _____

Estimate: about _____

f) 9 × 4,160

Rounded factor: _____

Multiply: _____ × _____ = _____

Estimate: about _____

PRACTICE Word problem

7. A class needs 7 packs of markers. Each pack costs $28. About how much will the markers cost? Is $250 enough?



$28 rounds to _____. 7 × _____ = _____

The markers cost about $_____. Is $250 enough? YES NO

How do you know?

_____

_____

MORE PRACTICE Skill builder



Round the big number to its greatest place. Multiply. Write your estimate.

a) 6 × 38
Rounded: _____
About _____

b) 4 × 73
Rounded: _____
About _____

c) 8 × 26
Rounded: _____
About _____

d) 5 × 451
Rounded: _____
About _____

e) 7 × 619
Rounded: _____
About _____

f) 3 × 864
Rounded: _____
About _____

g) 2 × 3,480
Rounded: _____
About _____

h) 6 × 7,250
Rounded: _____
About _____

i) 9 × 1,735
Rounded: _____
About _____

MORE PRACTICE Word problems



1. A store has 6 boxes of crayons. Each box holds 48 crayons. About how many crayons? Will the exact answer be more or less?

_____

_____

Answer: _____

2. A family drives 312 miles each day for 4 days. About how far do they drive? Will the exact answer be more or less?

_____

_____

Answer: _____

MIXED REVIEW Remember these?



From Lesson 1: 7 × 500

_____

= _____

From Lesson 1: 4 × 6,000

_____

= _____

CHECK YOURSELF **Show what you know**

1. Estimate 6 × 49

Rounded factor: _____

About _____

2. Estimate 4 × 812

Rounded factor: _____

About _____

Is the answer reasonable? Estimate first. Then circle YES or NO.

Answer to check	My estimate	Reasonable?
5 × 312 = 1,560	5 × _____ = _____	YES NO
3 × 690 = 1,270	3 × _____ = _____	YES NO

5. Find the mistake. Leo estimated 3 × 286 by rounding 3 to 0. He got 0. What should he do?

_____

_____

SAT CONNECTION

IF YOU MASTER
Estimating products

→ **IT HELPS YOU UNDERSTAND**
checking whether an answer makes sense

→ **ON THE SAT**
All four math domains: checking calculator results and ruling out answer choices that are far too big or too small

Example: 49 × 21 cannot be 10,290, because 50 × 20 is only 1,000

How did I do? Color one. I can do this alone · I need a little help · I need more practice

EDUCATOR TIP An estimate cannot prove an answer is right. It can show an answer is far off, like 1,270 for 3 × 690.

LESSON 3

Break It Apart: Partial Products with Arrays

Thursday, October 15

Name _____ Date _____

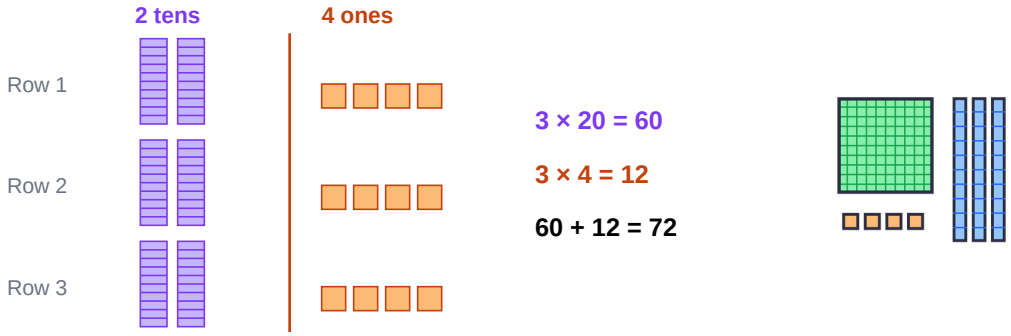
BIG IDEA Break the big number into tens and ones. Multiply each part. Then add the parts together.

WORDS TO KNOW

array	partial product	break apart
objects in equal rows	the answer for one part	split by place value

LEARN IT — 3 × 24 as an array

3 rows of 24. Each row has 2 tens and 4 ones.



STEPS

1. Break apart by place value. 24 = 20 + 4
2. Multiply each part. These are the partial products. 3 × 20 = 60 3 × 4 = 12
3. Add the partial products. 60 + 12 = **72**

Watch out: multiply the **value**, not the digit. In 24, the 2 means 20, so 3 × 20 = 60, not 6.

THREE DIGITS: ADD WITH REGROUPING. 7 × 286 = 1,400 + 560 + 42

Break apart: 286 = 200 + 80 + 6

Multiply each part: 7 × 200 = 1,400; 7 × 80 = 560; 7 × 6 = 42

	Th	H	T	O
	1	1		
	1	4	0	0
		5	6	0
+			4	2
	2	**0**	**0**	**2**

Line up the places. Add from the right.
Ones: 0 + 0 + 2 = 2.
Tens: 0 + 6 + 4 = 10 tens. Write 0, regroup 1 hundred.
Hundreds: 1 + 4 + 5 = 10 hundreds. Write 0, regroup 1 thousand.
Thousands: 1 + 1 = 2. 7 × 286 = **2,002**

TRY IT TOGETHER

Step by step

1. 2 × 34

34 = 30 + 4

2 × 30 = _____

2 × 4 = _____

Add: _____ + _____ = _____

2. 4 × 26

26 = _____ + _____

4 × _____ = _____

4 × _____ = _____

Add: _____ + _____ = _____

3. 3 × 142

142 = 100 + 40 + 2

Hundreds: 3 × 100 = _____

Tens: 3 × 40 = _____

Ones: 3 × 2 = _____

Add: _____ + _____ + _____ = _____

4. 5 × 213

213 = _____ + _____ + _____

5 × _____ = _____

5 × _____ = _____

5 × _____ = _____

Add: _____

5. 6 × 37 (fewer hints)

Estimate: about _____

Partial products: _____ _____

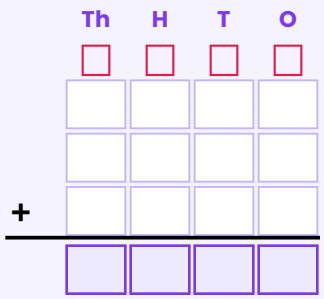
Add: _____ **Answer:** _____

6. 8 × 136 (add with regrouping)

8 × 100 = _____

8 × 30 = _____

8 × 6 = _____



Write the partial products in the boxes. Add from the right. Use the red boxes for regrouping.

Answer: _____

EDUCATOR TIP If place-value blocks are available, build problem 1 first. Otherwise, draw sticks for tens and dots for ones.

PRACTICE On your own

Estimate. Break apart. Multiply each part. Add.

a) 3 × 27

Estimate: about _____

Partial products:

_____

_____

Add: _____

Answer: _____

b) 6 × 43

Estimate: about _____

Partial products:

_____

_____

Add: _____

Answer: _____

c) 4 × 215

Estimate: about _____

Partial products:

_____

_____

_____

Add: _____

Answer: _____

d) 7 × 132

Estimate: about _____

Partial products:

_____

_____

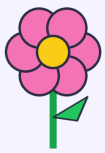
_____

Add: _____

Answer: _____

PRACTICE Word problem

5. A garden has 6 rows of tulips. Each row has 34 tulips. How many tulips are in the garden?



Estimate: 6 × _____ = _____

6 × _____ = _____ 6 × _____ = _____ _____ + _____ = _____

Answer: _____ tulips

MORE PRACTICE Skill builder

Estimate. Break apart. Multiply each part. Add.

a) 3 × 42

Estimate: about _____

_____

_____

_____

Answer: _____

b) 5 × 36

Estimate: about _____

_____

_____

_____

Answer: _____

c) 4 × 57

Estimate: about _____

_____

_____

_____

Answer: _____

d) 2 × 314

Estimate: about _____

_____

_____

_____

Answer: _____

e) 6 × 123

Estimate: about _____

_____

_____

_____

Answer: _____

f) 3 × 208

Estimate: about _____

_____

_____

_____

Answer: _____

MORE PRACTICE Word problems

1. 7 boxes each hold 46 markers. How many markers?

_____

_____

Answer: _____ markers

2. A school has 4 grades. Each grade has 128 students. How many students?

_____

_____

Answer: _____ students

MIXED REVIEW Remember these?

From Lesson 2: Estimate 6 × 52

_____

= _____

From Lesson 1: 8 × 400

_____

= _____

CHECK YOURSELF **Show what you know**

1. 4 × 23

Estimate: about _____

Partial products:

_____

_____

Add: _____

Answer: _____

2. 3 × 231

Estimate: about _____

Partial products:

_____

_____

_____

Add: _____

Answer: _____

3. Find the mistake. Ben wrote: 3 × 42 = 3 × 4 + 3 × 2 = 12 + 6 = 18.

The correct answer is _____.

What did Ben do wrong?

_____

_____

SAT CONNECTION

IF YOU MASTER
Breaking a number apart and multiplying each part

→ **IT HELPS YOU UNDERSTAND**
the distributive property

→ **ON THE SAT**
Advanced Math: equivalent expressions

Example: 3(x + 4) = 3x + 12, just like 3 × 24 = 3 × 20 + 3 × 4

How did I do? Color one. ○ I can do this alone ○ I need a little help ○ I need more practice

EDUCATOR TIP Check that the student writes one line for each place. A three-digit number needs three partial products.

LESSON 4

Rectangle Power: Area Models

Friday, October 16

Name _____ Date _____

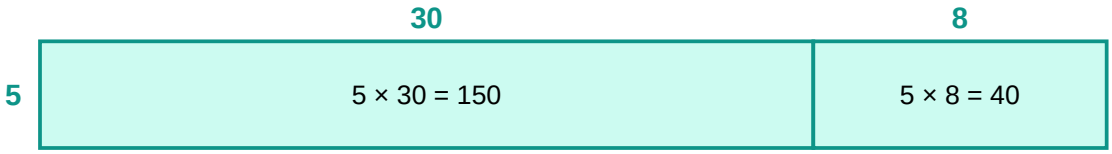
BIG IDEA A multiplication problem can be a rectangle. Split the rectangle by place value, find each small area, then add.

WORDS TO KNOW

area the space inside a shape	**area model** a rectangle split into parts	**label** write the value of each part

LEARN IT — 5 × 38 as a rectangle

The rectangle is 5 units tall and 38 units long. Split 38 into 30 + 8.



150 + 40 = 190

STEPS

1. Estimate. 5 × 38 is about 5 × 40 = 200.
2. Split the long side by place value. Write each part on top. 30 and 8
3. Write each part's product inside its rectangle. 150 and 40
4. Add. 150 + 40 = **190**. A little less than 200, because 38 was rounded up.

NOT TO SCALE The rectangles do not have to be the right size. Use the numbers you write on top, not the widths.

WATCH OUT! Label each part with its value. Write 30, not 3.

TRY IT TOGETHER — Step by step

1. 3 × 26. The labels are done. Fill in each rectangle.

	20	6
3	3 × 20 = _____	3 × 6 = _____

_____ + _____ = _____

2. 4 × 57. Write the missing label, then fill in each rectangle.

	50	_____
4	4 × 50 = _____	4 × ___ = _____

_____ + _____ = _____

3. 2 × 345. Three digits, three rectangles.

	300	40	5
2	2 × 300 = _____	2 × 40 = _____	2 × 5 = ____

_____ + _____ + _____ = _____

4. 6 × 128. Estimate: about _____. Write the labels on top and the products inside.

	_____	_____	_____
6	_____	_____	_____

_____ + _____ + _____ = _____

EDUCATOR TIP Students can trace each rectangle with a different colored pencil to match it with its partial product.

PRACTICE On your own

Estimate. **Write each place-value part above its rectangle. Write its partial product inside.** Then add.

a) 7 × 36

Estimate: about _____

_____ _____

7 | _____ | _____

Add: _____

Answer: _____

b) 5 × 84

Estimate: about _____

_____ _____

5 | _____ | _____

Add: _____

Answer: _____

c) 3 × 246

Estimate: about _____

_____ _____ _____

3 | _____ | _____ | _____

Add: _____

Answer: _____

d) 4 × 152

Estimate: about _____

_____ _____ _____

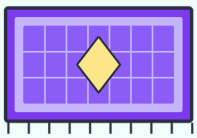
4 | _____ | _____ | _____

Add: _____

Answer: _____

PRACTICE Word problem

5. A classroom rug is 6 feet wide and 14 feet long. What is the area of the rug?



Estimate: 6 × _____ = _____

6 × 10 = _____ 6 × 4 = _____ _____ + _____ = _____

Answer: _____ square feet

MORE PRACTICE — Skill builder

Estimate. Write each part above its rectangle and its product inside. Then add.

a) 6 × 27

Estimate: about _____



Add: _____

Answer: _____

b) 9 × 43

Estimate: about _____

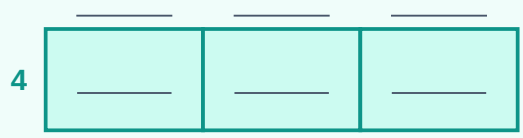


Add: _____

Answer: _____

c) 4 × 316

Estimate: about _____

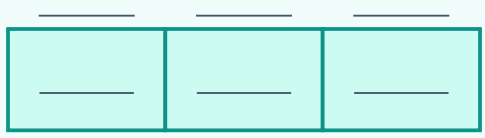


Add: _____

Answer: _____

d) 5 × 124

Estimate: about _____



Add: _____

Answer: _____

MORE PRACTICE — Word problems

1. A garden is 7 feet wide and 23 feet long. What is its area?

_____

_____

Answer: _____ square feet

2. A hallway rug is 3 feet wide and 145 feet long. What is its area?

_____

_____

Answer: _____ square feet

MIXED REVIEW — Remember these?

From Lesson 3: 5 × 38

_____

= _____

From Lesson 2: Estimate 4 × 688

_____

= _____

CHECK YOURSELF **Show what you know**

1. 8 × 24

Estimate: about _____

	_____	_____
8	_____	_____

Add: _____

Answer: _____

2. 3 × 152

Estimate: about _____

	_____	_____	_____
3	_____	_____	_____

Add: _____

Answer: _____

3. Find the mistake. Ava labeled her area model for 4 × 63 with 6 and 3. She got 24 + 12 = 36.

The correct labels are _____ and _____. The correct answer is _____.

Explain Ava's mistake:

_____

_____

SAT CONNECTION

IF YOU MASTER
Area models

→

IT HELPS YOU UNDERSTAND
the box method for multiplying expressions

→

ON THE SAT
Advanced Math: multiplying polynomials. Geometry: area

Example: $(x + 2)(x + 4) = x^2 + 6x + 8$, using the same split rectangle

How did I do? Color one. ○ I can do this alone ○ I need a little help ○ I need more practice

EDUCATOR TIP Area models and partial products are the same math. If the student prefers lines of partial products, accept them.

LESSON 5

Bigger Numbers: Four-Digit Factors

Monday, October 19

Name _____ Date _____

BIG IDEA Four digits means four places, so there are four partial products. The steps do not change.

WORDS TO KNOW

thousands the fourth place from the right	**zero place** a place with the digit 0

LEARN IT 2 × 3,214, one line for each place

Place	Value	Multiply	Partial product
Thousands	3,000	2 × 3,000	6,000
Hundreds	200	2 × 200	400
Tens	10	2 × 10	20
Ones	4	2 × 4	8

Add: 6,000 + 400 + 20 + 8 = **6,428**. Estimate: 2 × 3,000 = 6,000. 6,428 is a little more, because 3,214 was rounded down.

STEPS

1. Estimate.
2. Write one line for each place: thousands, hundreds, tens, ones.
3. Multiply each value.
4. Add the partial products.

A ZERO PLACE 3 × 2,031: the hundreds digit is 0, so 3 × 0 hundreds = 0. You may write it or skip it.

6,000 + 0 + 90 + 3 = **6,093**

WATCH OUT! Keep every place value. 2 × 300 is 600, not 6.

TRY IT TOGETHER **Step by step**

WORKED EXAMPLE: ADD WITH REGROUPING. 6 × 1,287 = 6,000 + 1,200 + 480 + 42

Break apart: 1,287 = 1,000 + 200 + 80 + 7

Multiply each part: 6 × 1,000 = 6,000; 6 × 200 = 1,200; 6 × 80 = 480; 6 × 7 = 42

	Th	H	T	O
	□	1	□	□
	6	0	0	0
	1	2	0	0
		4	8	0
+			4	2
	7	**7**	**2**	**2**

Ones: 0 + 0 + 0 + 2 = 2.
Tens: 0 + 0 + 8 + 4 = 12 tens. Write 2, regroup 1 hundred.
Hundreds: 1 + 0 + 2 + 4 = 7.
Thousands: 6 + 1 = 7. 6 × 1,287 = **7,722**

1. 2 × 1,342

Thousands: 2 × 1,000 = _____

Hundreds: 2 × 300 = _____

Tens: 2 × 40 = _____

Ones: 2 × 2 = _____

Add: _____

2. 3 × 2,123

Thousands: _____ × _____ = _____

Hundreds: _____ × _____ = _____

Tens: _____ × _____ = _____

Ones: _____ × _____ = _____

Add: _____

3. 4 × 1,212

Estimate: about _____

4 × _____ = _____

4 × _____ = _____

4 × _____ = _____

4 × _____ = _____

Add: _____

4. 3 × 2,031 (zero place)

Estimate: about _____

The hundreds digit is 0.

3 × 2,000 = _____

3 × 30 = _____

3 × 1 = _____

Add: _____

5. 4 × 1,276 (add with regrouping)

4 × 1,000 = _____ 4 × 200 = _____

4 × 70 = _____ 4 × 6 = _____

Write the partial products in the boxes. Add from the right. Use the red boxes for regrouping.

	Th	H	T	O
	□	□	□	□
+				

Answer: _____

PRACTICE On your own

Estimate. Break apart. Multiply each part. Add.

a) 2 × 4,321

Estimate: about _____

Partial products:

_____

_____

_____

_____

Add: _____

Answer: _____

b) 3 × 2,312

Estimate: about _____

Partial products:

_____

_____

_____

_____

Add: _____

Answer: _____

c) 5 × 1,104

Estimate: about _____

Partial products:

_____

_____

_____

_____

Add: _____

Answer: _____

d) 4 × 2,105

Estimate: about _____

Partial products:

_____

_____

_____

_____

Add: _____

Answer: _____

PRACTICE Word problem

5. A delivery truck carries 1,250 bottles of water. How many bottles do 3 trucks carry?



Estimate: 3 × _____ = _____

3 × 1,000 = _____ 3 × 200 = _____ 3 × 50 = _____

Add: _____ **Answer:** _____ bottles

MORE PRACTICE Skill builder

Estimate. Break apart. Multiply each part. Add.

a) 2 × 2,413

Estimate: about _____

_____

_____

_____

_____

Answer: _____

b) 3 × 1,322

Estimate: about _____

_____

_____

_____

_____

Answer: _____

c) 4 × 1,021

Estimate: about _____

_____

_____

_____

_____

Answer: _____

d) 5 × 1,203

Estimate: about _____

_____

_____

_____

_____

Answer: _____

e) 2 × 3,045

Estimate: about _____

_____

_____

_____

_____

Answer: _____

f) 3 × 2,201

Estimate: about _____

_____

_____

_____

_____

Answer: _____

MORE PRACTICE Word problems

1. 3 trains each carry 1,215 people. How many people is that?

_____

_____

Answer: _____ people

2. A library lends 2,350 books each month. How many books in 3 months?

_____

_____

Answer: _____ books

MIXED REVIEW Remember these?

From Lesson 3: 4 × 56

_____

= _____

From Lesson 1: 3 × 7,000

_____

= _____

CHECK YOURSELF **Show what you know**

1. 2 × 3,412

Estimate: about _____

Partial products:

_____

_____

_____

_____

Add: _____

Answer: _____

2. 4 × 1,201

Estimate: about _____

Partial products:

_____

_____

_____

_____

Add: _____

Answer: _____

3. Find the mistake. Noah wrote 2 × 1,423 = 2,000 + 8 + 4 + 6 = 2,018.

The correct answer is _____.

What did Noah forget?

_____

_____

SAT CONNECTION

IF YOU MASTER
Multiplying every place of a big number

→ **IT HELPS YOU UNDERSTAND**
distributing across many terms

→ **ON THE SAT**
Advanced Math: multiplying an expression with several terms

Example: $2(3x^2 + x + 4) = 6x^2 + 2x + 8$

How did I do? Color one. ○ I can do this alone ○ I need a little help ○ I need more practice

EDUCATOR TIP This is a stretch lesson in Virginia classrooms. Students who are not ready can skip it and return after Lesson 8.

LESSON 6

Math in Your Head

Tuesday, October 20

Name _____ Date _____

BIG IDEA Some numbers are easy to multiply in your head. Look for friendly pairs, or change a number to a friendly one.

WORDS TO KNOW

friendly number easy to multiply, like 100 or 200	**friendly pair** 4 × 25 = 100 or 2 × 50 = 100	**adjust** fix the extra

LEARN IT — Two mental math tools

Tool 1: Friendly Pairs

Find two numbers that make 100. Multiply them first.

You can change the order and grouping of factors without changing the product.

25 × 7 × 4

= (25 × 4) × 7

= 100 × 7

= **700**

Friendly pairs: 4 × 25, 2 × 50, 5 × 20

Tool 2: Make It Friendly

Change a number to a friendly one. Then fix the extra.

5 × 199

1. 199 is close to 200.
2. 5 × 200 = 1,000
3. 200 is 1 more than 199.
4. 5 × 1 = 5 extra
5. Subtract: 1,000 − 5 = **995**

ADD BACK: WHEN THE FRIENDLY NUMBER IS SMALLER

3 × 202

1. 202 is close to 200. 2. 3 × 200 = 600 3. 200 is 2 **less** than 202.
4. 3 × 2 = 6 is missing. 5. Add it back: 600 + 6 = **606**

Friendly number bigger than yours? **Subtract** the extra. Smaller? **Add back** what is missing.

WATCH OUT! Multiply the change by the other factor. In 5 × 199, subtract 5 × 1 = 5, not just 1.

TRY IT TOGETHER Tool 1: Friendly Pairs

1. 4 × 6 × 25

Friendly pair: 4 × 25 = _____

_____ × 6 = _____

2. 2 × 8 × 50

Friendly pair: _____ × _____ = 100

100 × _____ = _____

TRY IT TOGETHER Tool 2: Make It Friendly

3. 3 × 99 (subtract)

Friendly number: 100

3 × 100 = _____

Change: 100 − 99 = _____

Extra: 3 × _____ = _____

Subtract: _____ − _____ = _____

4. 4 × 202 (add)

Friendly number: 200

4 × 200 = _____

Change: 202 − 200 = _____

Missing: 4 × _____ = _____

Add: _____ + _____ = _____

5. 6 × 98 (fewer hints)

Friendly number: _____ 6 × _____ = _____

Add or subtract? _____ How much? _____ **Answer:** _____

EDUCATOR TIP Let the student say the steps aloud. Mental math still needs a written record of the extra, at least at first.

PRACTICE On your own. The tool is named for you.

a) 50 × 9 × 2

Use friendly pairs.

_____

_____

Answer: _____

b) 4 × 13 × 25

Use friendly pairs.

_____

_____

Answer: _____

c) 7 × 301

Use make it friendly.

_____

_____

Answer: _____

d) 5 × 298

Use make it friendly.

_____

_____

Answer: _____

PRACTICE Word problem

5. 4 teams each fill 25 water bottles a day for 6 days. How many bottles do they fill? **4 × 25 × 6**

Friendly pair: 4 × 25 = _____ _____ × 6 = _____

Answer: _____ bottles



MORE PRACTICE Skill builder

Use the tool named. Write the extra step if you need it.

a) 25 × 9 × 4

friendly pairs

_____

Answer: _____

b) 2 × 17 × 50

friendly pairs

_____

Answer: _____

c) 4 × 8 × 25

friendly pairs

_____

Answer: _____

d) 4 × 99

make it friendly

_____

Answer: _____

e) 6 × 199

make it friendly

_____

Answer: _____

f) 3 × 301

make it friendly

_____

Answer: _____

g) 8 × 102

make it friendly

_____

Answer: _____

h) 5 × 98

make it friendly

_____

Answer: _____

i) 50 × 6 × 2

friendly pairs

_____

Answer: _____

MORE PRACTICE Word problems

1. 3 classes each read 99 pages. How many pages in all? Use make it friendly.

_____

_____

Answer: _____ pages

2. Each of 7 classrooms has 4 boxes of 25 pencils. How many pencils? Use friendly pairs.

_____

_____

Answer: _____ pencils

MIXED REVIEW Remember these?

From Lesson 3: 3 × 241

_____

= _____

From Lesson 2: Estimate 7 × 289

_____

= _____

CHECK YOURSELF Show what you know

1. 25 × 3 × 4

_____

= _____

2. 5 × 299

_____

= _____

3. 3 × 402

_____

= _____

4. Find the mistake. Lina found 6 × 99 by doing 600 − 1 = 599.

The correct answer is _____.

What should Lina subtract, and why?

_____

_____

SAT CONNECTION

IF YOU MASTER
Friendly pairs and making numbers friendly

→

IT HELPS YOU UNDERSTAND
properties of operations and rewriting expressions

→

ON THE SAT
Algebra and Advanced Math: rewriting expressions to make them easier to work with

Example: 5 × 199 = 5(200 − 1), the same idea as 5(x − 1)

How did I do? Color one. ○ I can do this alone ○ I need a little help ○ I need more practice

EDUCATOR TIP Students who struggle with Make It Friendly can always fall back on breaking apart: 6 × 99 = 540 + 54.

LESSON 7

Pick Your Tool: Choose a Strategy

Wednesday, October 21

Name _____ Date _____

BIG IDEA Look at the numbers before you multiply. Pick the tool that makes the numbers easiest. Any correct tool is fine.

LEARN IT — Which tool should I use?

Look at the numbers...	Use this tool	Example
One number ends in zeros	Zero pattern	6 × 400 = 2,400
Close to a friendly number (98, 199, 302)	Make it friendly	4 × 98 = 400 − 8 = 392
A friendly pair (4 × 25, 2 × 50)	Friendly pairs	4 × 7 × 25 = 100 × 7 = 700
None of these	Break it apart or area model	3 × 214 = 600 + 30 + 12 = 642

WORKED EXAMPLE: 5 × 198

1. Look: 198 is close to 200. I will make it friendly.
2. 5 × 200 = 1,000
3. 200 is 2 more than 198. 5 × 2 = 10 extra.
4. Subtract: 1,000 − 10 = **990**

Another way: break apart. 5 × 100 + 5 × 90 + 5 × 8 = 500 + 450 + 40 = 990. Same answer!

TOOLBOX TIP Breaking apart always works. The other tools are shortcuts for special numbers.

TRY IT TOGETHER — Step by step

1. 3 × 199

199 is close to _____. I will **make it friendly**.

3 × _____ = _____

Extra: 3 × _____ = _____

Subtract: _____ − _____ = _____

2. 3 × 312

Circle one: **break apart** **area model**

312 = _____

_____

_____

_____

Answer: _____

3. Name the best tool. Write: zero pattern, make it friendly, friendly pairs, or break apart.

8 × 300	25 × 6 × 4	7 × 99	4 × 236
_____	_____	_____	_____

4. Pick your own tool: 6 × 205

Tool: _____

_____

_____

Answer: _____

EDUCATOR TIP Ask "Why did you choose that tool?" more often than "Is it right?" Any correct strategy earns full credit.

PRACTICE On your own

Estimate. Choose a tool. Show your work.

a) 6 × 49

Estimate: about _____

Tool: _____

_____

_____

_____

Answer: _____

b) 4 × 250

Estimate: about _____

Tool: _____

_____

_____

_____

Answer: _____

c) 8 × 125

Estimate: about _____

Tool: _____

_____

_____

_____

Answer: _____

d) 7 × 402

Estimate: about _____

Tool: _____

_____

_____

_____

Answer: _____

PRACTICE Word problem

5. A school buys 3 boxes of markers for $48 each and 2 boxes of paper for $25 each. What is the total cost?



Markers: 3 × $48 = _____ Paper: 2 × $25 = _____

Total: _____ + _____ = _____

MORE PRACTICE Skill builder

Choose a tool. Show your work.

a) 5 × 600

Tool: _____

_____

_____

Answer: _____

b) 4 × 199

Tool: _____

_____

_____

Answer: _____

c) 25 × 8 × 4

Tool: _____

_____

_____

Answer: _____

d) 3 × 156

Tool: _____

_____

_____

Answer: _____

e) 7 × 98

Tool: _____

_____

_____

Answer: _____

f) 6 × 234

Tool: _____

_____

_____

Answer: _____

MORE PRACTICE Word problems

1. A teacher buys 4 packs of paper. Each pack costs $99. What is the total cost?

_____

_____

Answer: _____

2. A cafe packs 5 boxes with 125 cups each. How many cups?

_____

_____

Answer: _____ cups

MIXED REVIEW Remember these?

From Lesson 5: 2 × 1,302

_____

= _____

From Lesson 4: 6 × 45

_____

= _____

CHECK YOURSELF **Show what you know**

1. 9 × 99

Estimate: about _____

Tool: _____

_____

_____

_____

Answer: _____

2. 6 × 215

Estimate: about _____

Tool: _____

_____

_____

_____

Answer: _____

3. Two ways. Find 4 × 102 two different ways.

Way 1:

_____

Way 2:

_____

Both ways give _____.

SAT CONNECTION

IF YOU MASTER
Choosing a strategy

→

IT HELPS YOU UNDERSTAND
picking the fastest correct method

→

ON THE SAT
All four math domains: choosing between an equation, a graph, a table, or mental math

Example: some questions are quickest with algebra, others with the built-in graphing calculator

How did I do? Color one.

○ I can do this alone ○ I need a little help ○ I need more practice

EDUCATOR TIP Solving one problem two ways is a strong check. If the answers differ, the student knows to look for a mistake.

LESSON 8

Draw It, Solve It: Bar Models

Thursday, October 22

Name _____ Date _____

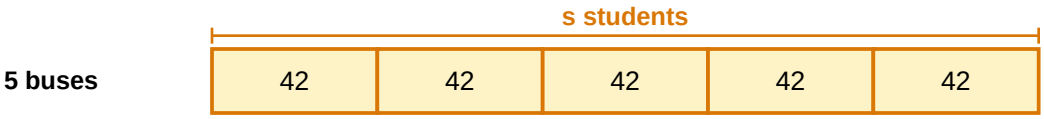
BIG IDEA A bar model shows how the numbers in a story are related. Draw one box for each equal group.

WORDS TO KNOW

bar model boxes that show the numbers	**equal groups** groups of the same size	**times as many** a comparison by multiplying

LEARN IT Equal groups

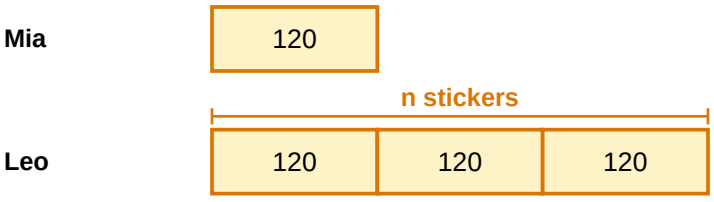
5 buses each carry 42 students. How many students, **s**, are there?



s = 5 × 42 **s** = 210 students

LEARN IT Times as many

Mia has 120 stickers. Leo has 3 times as many. How many stickers, **n**, does Leo have?



n = 3 × 120 **n** = 360 stickers

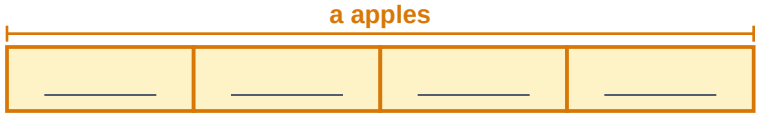
STEPS

1. Read. What do I need to find?
2. Draw a box for each group. Write the number in each box.
3. Write an equation with a letter for the unknown.
4. Solve. Check with an estimate.

TRY IT TOGETHER

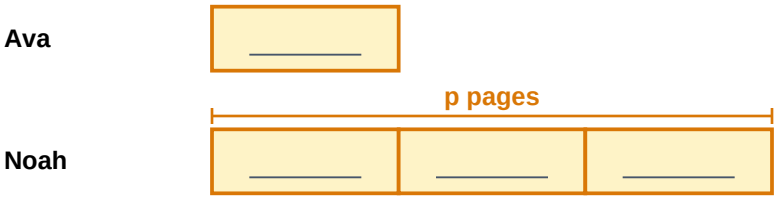
Step by step

1. Finish the model. 4 baskets each hold 36 apples. How many apples, **a**?



a = _____ × _____ **a** = _____ apples

2. Times as many. Ava read 135 pages. Noah read 3 times as many. How many pages, **p**, did Noah read?

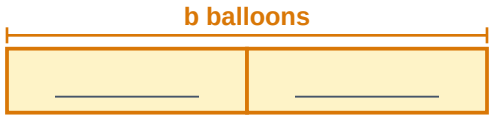


p = _____ × _____ **p** = _____ pages

3. Two steps. A shop sold 114 red balloons and 128 blue balloons on Monday. On Tuesday it sold 2 times as many. How many balloons, **b**, on Tuesday?

Step 1: Monday's total: 114 + 128 = _____

Step 2: Write Monday's total in each box.



b = 2 × _____ **b** = _____ balloons

EDUCATOR TIP Point from each number in the equation to the matching part of the bar model. They should match.

PRACTICE On your own. Draw a bar model, write an equation, and solve.

a) 6 boxes each hold 48 crayons. How many crayons, c?

Bar model:

Equation: _____

Answer: _____ crayons

b) A library has 7 shelves. Each shelf holds 64 books. How many books, k?

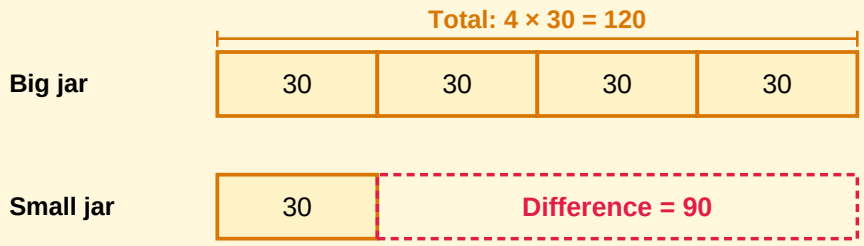
Bar model:

Equation: _____

Answer: _____ books

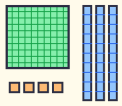
PRACTICE How much more?

EXAMPLE A small jar contains 30 beads. A big jar contains four times as many. How many more beads are in the big jar than in the small jar?



Total: 120. **One group:** 30. **Difference:** 120 − 30 = **90** beads.

c) A small bin contains 140 blocks. A big bin contains five times as many. How many more blocks are in the big bin than in the small bin?



Total: 5 × 140 = _____

Difference: _____ − 140 = _____

Answer: _____ more blocks

MORE PRACTICE Skill builder

Draw a bar model. Write an equation. Solve.

a) 5 shelves each hold 36 books. How many books, b?

Equation: _____

Answer: _____

b) Lily ran 125 meters. Her brother ran 4 times as far. How far, d, did her brother run?

Equation: _____

Answer: _____

c) Sam has 85 stamps. Ann has 3 times as many. How many stamps do they have together, t?

Equation: _____

Answer: _____

d) A small tank contains 150 liters of water. A big tank contains six times as much. How many more liters are in the big tank than in the small tank?

Equation: _____

Answer: _____

MIXED REVIEW Remember these?

From Lesson 6: 4 × 99

_____

= _____

From Lesson 3: 3 × 215

_____

= _____

CHECK YOURSELF **Show what you know**

1. 8 teams have 15 players each. How many players, t? Draw a bar model.

Equation: _____

Answer: _____ players

2. Zara has 145 cards. Her brother has 4 times as many. How many cards do they have together, c?

Equation(s): _____

Answer: _____ cards

SAT CONNECTION

IF YOU MASTER
Bar models with a letter for the unknown

→

IT HELPS YOU UNDERSTAND
writing an equation from a story

→

ON THE SAT
Algebra: linear equations in one variable. Problem-Solving and Data Analysis: ratios ("times as many")

Example: "Leo has 3 times as many as Mia, who has 120" becomes n = 3 × 120

How did I do? Color one. ○ I can do this alone ○ I need a little help ○ I need more practice

You finished the book!

You can now multiply numbers with up to four digits by a one-digit number, estimate, and choose a strategy. Look back at Lesson 1. How much have you grown?

EDUCATOR TIP Problem 2 has two steps. Accept 145 × 5 = 725 or 4 × 145 = 580, then 145 + 580 = 725.

Answer Key

Accept other reasonable estimates and any correct strategy that is shown.

Lesson 1

Try It Together	1. 12; 1 zero; 120. 2. 6 × 5 = 30; 2 zeros; 3,000. 3. 21, 210, 2,100, 21,000; 30, 300, 3,000, 30,000. 4. 8 × 2 = 16; 3 zeros; 16,000. 5. 4 × 5 = 20; YES; 2,000. 6. 630. 7. 18,000. 8. 300.
Practice	a) 240 b) 2,100 c) 18,000 d) 4,000 e) 320 f) 42,000 g) 20,000 h) 5,400. 9. 7 × 2 = 14; 2 zeros; 1,400 pencils. 10. 4 × 3,000 = 12,000 muffins. 11. 10 × 100 = 1,000. No; this product ends in three zeros.
More Practice	a) 150 b) 5,600 c) 24,000 d) 180 e) 4,200 f) 1,000 g) 27,000 h) 3,200 i) 480 j) 5,400 k) 20,000 l) 21,000 Word problem 1: 1,800 apples. Word problem 2: 360 miles.
Check Yourself	1. 360 2. 2,400 3. 30,000. 4. 2,000. 4 × 5 = 20, so 4 × 500 is 20 hundreds, or 2,000. Sam's answer is ten times too small. 5. Sample answers: 8 × 300, 6 × 400, 3 × 800.

Lesson 2

Try It Together	1. 60. 2. 60; 3 × 60 = 180; about 180. 3. 700; 6 × 700 = 4,200; about 4,200. 4. thousands; 3,000; 5 × 3,000 = 15,000; about 15,000. 5. 40; 7 × 40 = 280; about 280; DOWN. 6. MORE. Sample: 41 was rounded down to 40, so the exact answer (287) is more than 280.
Practice	a) 40; 7 × 40 = 280 b) 70; 4 × 70 = 280 c) 300; 8 × 300 = 2,400 d) 600; 3 × 600 = 1,800 e) 2,000; 6 × 2,000 = 12,000 f) 4,000; 9 × 4,000 = 36,000. 7. $30; 7 × 30 = 210; about $210; YES. Sample: $28 was rounded up, so the real cost is less than $210, and $210 is less than $250. (Exact: $196.)
More Practice	a) 40; 6 × 40 = 240. b) 70; 4 × 70 = 280. c) 30; 8 × 30 = 240. d) 500; 5 × 500 = 2,500. e) 600; 7 × 600 = 4,200. f) 900; 3 × 900 = 2,700. g) 3,000; 2 × 3,000 = 6,000. h) 7,000; 6 × 7,000 = 42,000. i) 2,000; 9 × 2,000 = 18,000. Word problem 1: 48 → 50; 6 × 50 = 300; about 300 crayons; less, because 48 was rounded up (exact 288). Word problem 2: 312 → 300; 4 × 300 = 1,200; about 1,200 miles; more, because 312 was rounded down (exact 1,248). Review: 7 × 500 = 3,500; 4 × 6,000 = 24,000.
Check Yourself	1. 50; about 300. 2. 800; about 3,200. 3. 5 × 300 = 1,500; YES (1,560 is a little more, because 312 was rounded down). 4. 3 × 700 = 2,100; NO (exact 2,070). 5. Sample: round the big number, not the 3. 286 rounds to 300, so about 900.

Lesson 3

Try It Together	1. 60; 8; 60 + 8 = 68. 2. 20 + 6; 80; 24; 80 + 24 = 104. 3. 300; 120; 6; 300 + 120 + 6 = 426. 4. 200 + 10 + 3; 1,000; 50; 15; 1,000 + 50 + 15 = 1,065. 5. About 240 (6 × 40); 180 and 42; 180 + 42 = 222. 6. 800; 240; 48; 800 + 240 + 48 = 1,088 (tens: 4 + 4 = 8; hundreds: 8 + 2 = 10, write 0 and regroup 1 thousand).
Practice	Sample estimates: a) 90 b) 240 c) 800 d) 700. a) 60 + 21 = 81 b) 240 + 18 = 258 c) 800 + 40 + 20 = 860 d) 700 + 210 + 14 = 924. 5. Estimate 6 × 30 = 180; 6 × 30 = 180; 6 × 4 = 24; 180 + 24 = 204 tulips.
More Practice	a) est. 120; 120 + 6 = 126. b) est. 200; 150 + 30 = 180. c) est. 240; 200 + 28 = 228. d) est. 600; 600 + 20 + 8 = 628. e) est. 600; 600 + 120 + 18 = 738. f) est. 600; 600 + 24 = 624 (a 0 partial product may be written or skipped). Word problem 1: 322 markers. Word problem 2: 512 students. Review: Estimate 6 × 52: 50; about 300; 8 × 400 = 3,200.
Check Yourself	1. Estimate 80; 80 + 12 = 92. 2. Estimate 600; 600 + 90 + 3 = 693. 3. 126. Sample: Ben multiplied the digit 4 instead of its value 40. 3 × 40 = 120, plus 6, is 126.

Lesson 4

Try It Together	1. 60; 18; 60 + 18 = 78. 2. Label 7; 200; 28; 200 + 28 = 228. 3. 600; 80; 10; 600 + 80 + 10 = 690. 4. Estimate about 600 (6 × 100); labels 100, 20, 8; 600, 120, 48; 600 + 120 + 48 = 768.
Practice	Sample estimates: a) 280 b) 400 c) 600 d) 800. a) 30 | 6: 210 + 42 = 252 b) 80 | 4: 400 + 20 = 420 c) 200 | 40 | 6: 600 + 120 + 18 = 738 d) 100 | 50 | 2: 400 + 200 + 8 = 608. 5. Estimate 6 × 10 = 60; 60; 24; 60 + 24 = 84 square feet.
More Practice	a) est. 180; 120 + 42 = 162. b) est. 360; 360 + 27 = 387. c) est. 1,200; 1,200 + 40 + 24 = 1,264. d) est. 500; 500 + 100 + 20 = 620. Word problem 1: 161 square feet. Word problem 2: 435 square feet. Review: 5 × 38 = 190; Estimate 4 × 688: 700; about 2,800.
Check Yourself	1. Estimate 160; 20 | 4: 160 + 32 = 192. 2. Estimate 600 (152 rounds to 200); 100 | 50 | 2: 300 + 150 + 6 = 456. 3. Labels 60 and 3; 252. Sample: Ava wrote the digit 6 instead of its value 60, so her first product was 24 instead of 240.

Lesson 5

Try It Together	1. 2,000; 600; 80; 4; 2,000 + 600 + 80 + 4 = 2,684. 2. 3 × 2,000 = 6,000; 3 × 100 = 300; 3 × 20 = 60; 3 × 3 = 9; 6,369. 3. About 4,000; 4 × 1,000 = 4,000; 4 × 200 = 800; 4 × 10 = 40; 4 × 2 = 8; 4,848. 4. About 6,000; 6,000; 90; 3; 6,093. 5. 4,000; 800; 280; 24; 4,000 + 800 + 280 + 24 = 5,104 (tens: 8 + 2 = 10, write 0, regroup 1; hundreds: 1 + 8 + 2 = 11, write 1, regroup 1; thousands: 1 + 4 = 5).

Lesson 5

Practice	Sample estimates: a) 8,000 b) 6,000 c) 5,000 d) 8,000. a) 8,000 + 600 + 40 + 2 = 8,642 b) 6,000 + 900 + 30 + 6 = 6,936 c) 5,000 + 500 + 0 + 20 = 5,520 (the 0 tens partial product may be written or skipped) d) 8,000 + 400 + 0 + 20 = 8,420. 5. Estimate 3 × 1,000 = 3,000; 3,000; 600; 150; 3,750 bottles.
More Practice	a) est. 4,000; 4,000 + 800 + 20 + 6 = 4,826. b) est. 3,000; 3,000 + 900 + 60 + 6 = 3,966. c) est. 4,000; 4,000 + 80 + 4 = 4,084 (a 0 partial product may be written or skipped). d) est. 5,000; 5,000 + 1,000 + 15 = 6,015 (a 0 partial product may be written or skipped). e) est. 6,000; 6,000 + 80 + 10 = 6,090 (a 0 partial product may be written or skipped). f) est. 6,000; 6,000 + 600 + 3 = 6,603 (a 0 partial product may be written or skipped). Word problem 1: 3,645 people. Word problem 2: 7,050 books. Review: 4 × 56 = 224; 3 × 7,000 = 21,000.
Check Yourself	1. Estimate 6,000; 6,000 + 800 + 20 + 4 = 6,824. 2. Estimate 4,000; 4,000 + 800 + 0 + 4 = 4,804. 3. 2,846. Sample: Noah multiplied the digits instead of their values. 2 × 400 = 800, 2 × 20 = 40, 2 × 3 = 6.

Lesson 6

Try It Together	1. 100; 100 × 6 = 600. 2. 2 × 50 = 100; 100 × 8 = 800. 3. 300; 1; 3 × 1 = 3; 300 − 3 = 297. 4. 800; 2; 4 × 2 = 8; 800 + 8 = 808. 5. 100; 600; subtract; 12; 588.
Practice	a) 50 × 2 = 100; 100 × 9 = 900. b) 4 × 25 = 100; 100 × 13 = 1,300. c) 7 × 300 = 2,100; add 7 × 1 = 7; 2,107. d) 5 × 300 = 1,500; subtract 5 × 2 = 10; 1,490. 5. 100; 600 bottles.
More Practice	a) 900. b) 1,700. c) 800. d) 396. e) 1,194. f) 903. g) 816. h) 490. i) 600. Word problem 1: 297 pages. Word problem 2: 700 pencils. Review: 3 × 241 = 723; Estimate 7 × 289: 300; about 2,100.
Check Yourself	1. 300. 2. 1,495 (1,500 − 5). 3. 1,206 (1,200 + 6). 4. 594. Sample: Lina should subtract 6 × 1 = 6, because each of the 6 groups had 1 extra.

Lesson 7

Try It Together	1. 200; 600; 3 × 1 = 3; 600 − 3 = 597. 2. Either tool; 300 + 10 + 2; 900 + 30 + 6 = 936. 3. zero pattern; friendly pairs; make it friendly; break apart (an area model is also fine). 4. Sample: break apart, 1,200 + 30 = 1,230.
Practice	Estimates (greatest-place rounding): a) 300 b) 1,200 (250 rounds to 300; 4 × 300 = 1,200) c) 800 d) 2,800. For b, 1,000 is the exact answer; it may also be accepted as a compatible-number estimate if the student explains that approach. a) 294 (make it friendly: 300 − 6) b) 1,000 (break apart 800 + 200, or 4 × 25 = 100, so 4 × 250 = 1,000) c) 1,000 (800 + 200) d) 2,814 (2,800 + 14). Accept any correct tool. 5. $144; $50; $144 + $50 = $194.

Lesson 7

More Practice	a) 3,000. b) 796. c) 800. d) 468. e) 686. f) 1,404. Accept any correct tool. Word problem 1: $396. Word problem 2: 625 cups. Review: 2 × 1,302 = 2,604; 6 × 45 = 270.
Check Yourself	1. Estimate 900; 891 (900 − 9). 2. Estimate 1,200; 1,290 (1,200 + 60 + 30). 3. Partial products: 4 × 100 + 4 × 2 = 408. Compensation: 4 × 105 − 4 × 3 = 420 − 12 = 408. Both ways give 408.

Lesson 8

Try It Together	1. Four boxes of 36; a = 4 × 36; a = 144 apples. 2. Ava: one box of 135; Noah: three boxes of 135; p = 3 × 135; p = 405 pages. 3. 242; two boxes of 242; b = 2 × 242; b = 484 balloons.
Practice	a) Six boxes of 48; c = 6 × 48 = 288 crayons. b) Seven boxes of 64; k = 7 × 64 = 448 books. c) 700; 700 − 140 = 560; 560 more blocks.
More Practice	a) b = 5 × 36 = 180 books. b) d = 4 × 125 = 500 meters. c) t = 4 × 85 = 340 stamps (or 255 + 85). d) 6 × 150 = 900; 900 − 150 = 750 liters. Review: 4 × 99 = 396; 3 × 215 = 645.
Check Yourself	1. Eight boxes of 15; t = 8 × 15 = 120 players. 2. Zara one box of 145, brother four boxes of 145; c = 5 × 145 = 725 cards (or 4 × 145 = 580, then 145 + 580 = 725).